K-Nearest Neighbors

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Example: Chronic Kidney Disease

Doctors want to know whether their patients are at risk for developing chronic kidney disease (CKD). They have a dataset of many patients with measurements from a blood test, as well as whether the patients developed CKD. We want to develop a model to predict whether a patient will get CKD using just the information from the blood test.

Intuition

Intuition behind K-Nearest Neighbors: When predicting future data, we might want to consider how other similar observations might have performed in the past.

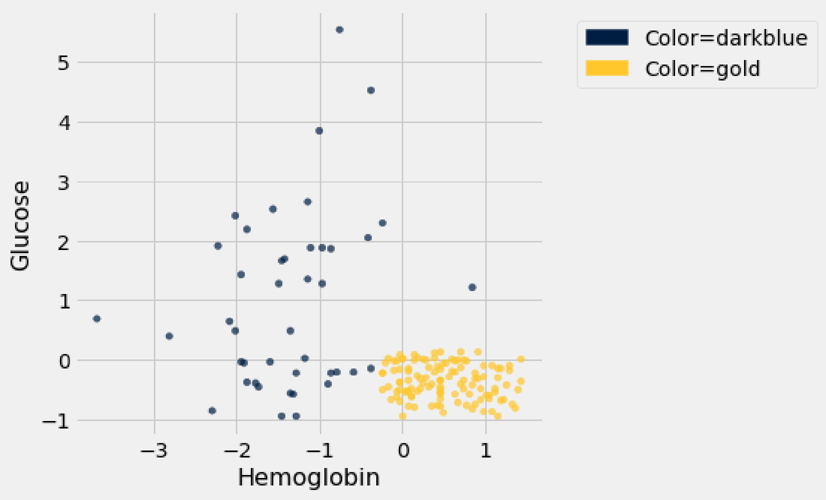
Example: When trying to predict whether a patient will develop chronic kidney disease, we might want to think about whether patients with similar blood test results got CKD.

K-Nearest Neighbors Algorithm

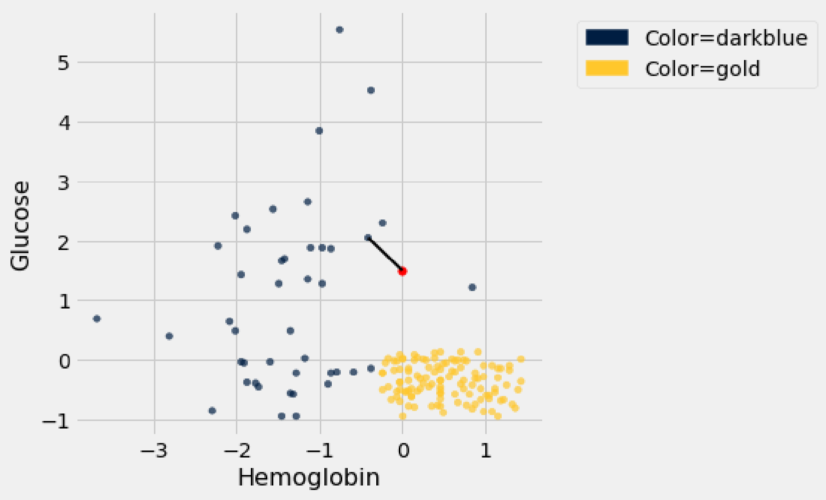
Find the data point that is closest to the one you want to predict.

The predicted class is the class of the nearest data point.

Data


Color=darkblue
Color=gold
Glucose
Hemoglobin
5
4
3
2
1
0
−1
−3
−2
−1
0
1

Prediction using K-Nearest Neighbors

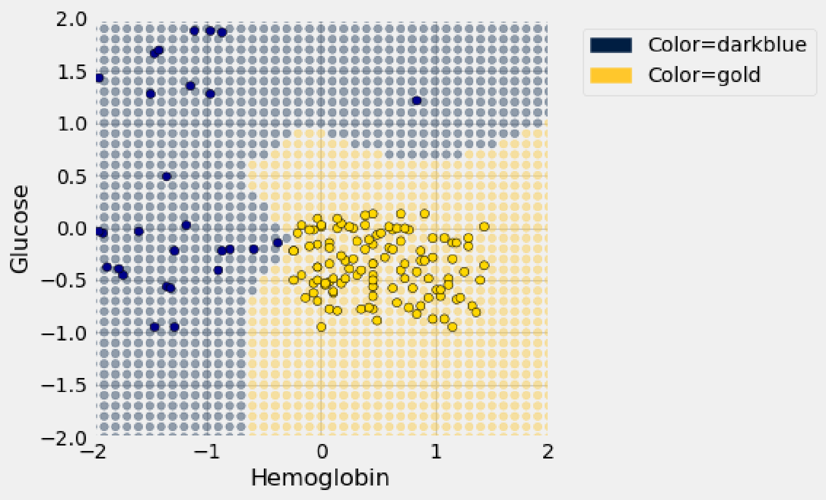


K-Nearest Neighbors

This shows how we might predict any new data point. We might be more interested in the **overall model**. That is, given any input, what would be our predicted output?

We can describe this model using a **decision boundary**.

The K-Nearest Neighbors Model

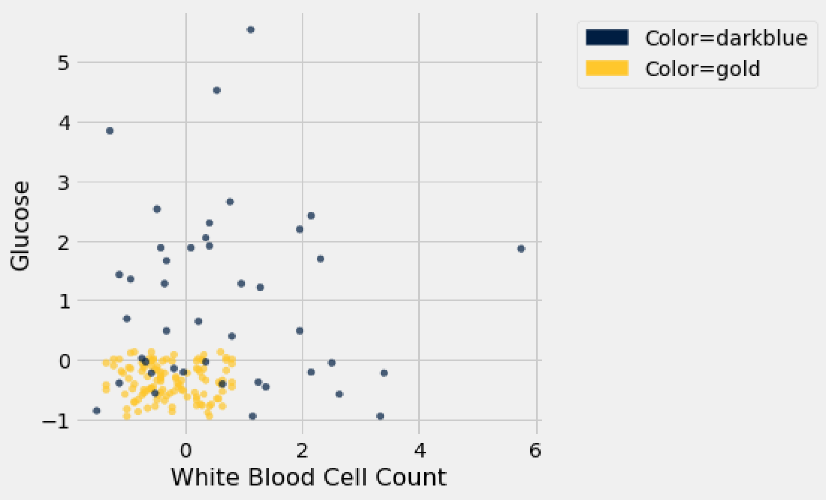


K-Nearest Neighbors

What we've described is actually 1-Nearest Neighbors. Usually, we choose a number of closest points and use those together to determine predicted class.

For example, we might set k = 5 and choose five closest points and predict using a "vote" of those classes.

Another Example



Considerations

Choosing K

- We want to choose K to minimize **overfitting**.
- K is arbitrary! You may want to try multiple values to see which performs best.
- Use an odd number (to avoid ties)